DATS6101 Project 2 Proposal
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MTA: Dataset Source
Census: Dataset Source

GitHub Link

For our second project, we are going to further build on the MTA Subway dataset that we performed EDA and inferential tests on in project 1. In the first, we identified significant differences for ridership between boroughs, time of day, weekdays versus weekends, and commuting periods. In project 2, we plan to investigate whether these changes can be used to predict station-level ridership. Further, we would like to include census data (median household income, population density) to be able to see whether incorporating census data would improve the ridership predictions.

Our primary question for this project is “Can we predict hourly ridership at NYC subway stations using time, day, and borough characteristics?”. We would also like to be able to predict and classify stations according to ridership patterns (high, medium, low) and identify which variables are most useful for identifying the categories.

We will be using the same dataset from Project 1 for New York City subway hourly ridership for the week of September 15-21, consisting of approximately 553,429 rows of 12 variables. To supplement the analysis, we will be using census information from the American Community Survey (ACS) 2024 dataset. This data will be pulled into R through an API key, and we will be able to get demographic statistics for each neighborhood that is included in our original dataset. (Note: this process is going to involve a lot of cleaning, so we are currently unsure of the exact number of observations).